

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain Double Pipe Heat Exchanger with Sketch in detail.
- Q.37 Explain plate and frame filter press with neat sketch.
- Q.38 Explain Carnot cycle and how its efficiency is calculated.

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Note: Multiple choice questions. All questions are compulsory (10x1=10)

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- Q.5 The efficiency of the ideal screens is _____
- a) Unity b) Infinity
c) Zero d) Can't be determined
- Q.6 The heat is transferred by conduction, convection and radiation simultaneously in
- a) Melting of ice b) Boiler furnaces
c) Condensation of steam in condenser
d) None of these
- Q.7 The first law of thermodynamics gives only the change on energy E for the process.
- a) True b) False
- Q.8 Which of the following is a type of coarse crusher?
- a) Rod mill b) Stamp battery
c) Jaw crusher d) Dicer
- Q.9 Which of the following works principle of compression?
- a) Rod mill b) Knife cutter
c) Blake jaw crusher d) Gyratory crusher
- Q.10 What is the unit of diffusion coefficient?
- a) m^2 b) s
c) m^2s d) m^2/s

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Heat transfer operation.
- Q.12 Define Energy.
- Q.13 Define Grinding.
- Q.14 Name various mode of Heat Transfer.
- Q.15 Define Fick's law
- Q.16 The value of C_p is always _____ than C_v .
- Q.17 Define the Carnot cycle.
- Q.18 Define isochoric process.
- Q.19 Define Conduction.
- Q.20 Define principle of filtration.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 State Kick's law and bonds in crushing.
- Q.22 Explain Fourier law of heat conduction for single conductive wall.
- Q.23 Define the principle of ball mill and give its use.
- Q.24 Define conduction and convection with suitable examples.

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